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Extracting Core Meaning from Legal Queries Using Semantic Technologies

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Abstract. Legal queries are often expressed in unstructured, ambiguous, or noisy natural language, which poses significant challenges for accurate information retrieval. This paper presents a comprehensive framework for legal question answering that integrates semantic technologies—including an ontology platform, knowledge graph, and large language models (LLMs)—to improve question understanding and response accuracy. We propose a multi-step pre-processing pipeline that standardizes legal questions using spelling correction, abbreviation expansion, syntactic restructuring, and semantic summarization supported by LLMs. Besides, a query system is designed that maps pre-processed questions into graph-based representations and performs subgraph matching over a legal knowledge graph. The system is evaluated on real-world legal questions collected from online forums covering traffic law and social insurance. The results show that the proposed approach achieves a high semantic similarity score (avg. cosine similarity of 0.9078 after standardization) and outperforms baseline LLMs like ChatGPT and Gemini in query accuracy (up to 82.25% in certain question categories). These findings highlight the effectiveness of combining LLMs and semantic structures for robust legal information retrieval.

Keywords. Legal Information, Legal Query System, Large Language Models, Information retrieval, Information Denoising

1. Introduction

In the current context, systems such as chatbots, virtual assistants and Question-Answering (QA) platforms have become popular and are being applied to meet many fields such as medicine, education and especially in the legal field [11,12], a field that is always very necessary for people and society [13,14]. This leads to an increasing demand for human-computer interaction as well as the need for intelligent queries. However, a big problem is that the questions that users often ask the systems are often inconsistent, prone to spelling mistakes or incorrect grammatical structure, which can make it very difficult

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to query information, and very easy to give wrong answers [15]. The knowledge-based system can deduce to achieve the correct results by using the human reasoning described as the knowledge core of the system [28].

In the legal field, when users' questions are often not structured correctly and provide a lot of additional confusing data, this is a big challenge for the process of querying legal information [26]. If the answer to the user is incorrect, it will cause very serious consequences.

Therefore, a solution is needed to answer a variety of questions posed by users as well as ensure high accuracy of the answers [19,20]. An important task that needs to be solved is to accurately identify the main content of the question as well as eliminate unnecessary information to help the system more easily understand the semantics of the question [21,27]. This is an important first step to help improve the system's ability to answer questions accurately and ensure good performance of the program.

This study presents a process for building a legal information query program that can handle a variety of questions posed by users based on a knowledge base organized in the form of a knowledge graph combined with a graph matching query method [22]. The novelty of the system is the solution to standardize the questions entered by the user to help the system understand the main idea of the question, avoid wrong answers due to noisy information, and improve the accuracy and performance of the system. This method is capable of supporting pre-processing questions to help correct spelling errors, standardize special characters, and summarize and condense the main ideas in the sentence. To do this, this study takes advantage of the context understanding ability of Large Language Models (LLMs) and the sentence analysis ability of natural language processing (NLP) tools to help shorten and restructure the questions in the simplest and most understandable way for query systems.

The next section will review related works in the field of intelligent query support and detailed studies on summarizing question content. Section 3 presents the detailed process of building the query system with a focus on pre-processing to standardize the questions as well as the reduction and restructuring of the questions. Experiments are conducted to determine the accuracy of the presented method as well as the experimental results with real questions in the field of road traffic law and social insurance law in Section 4. The last section concludes the main results of this paper and proposes the future research directions.

2. Related Work

Nowadays, the research to support users in asking and answering legal information as well as improving the accuracy in the systems is always of interest and development to best serve users [16,17]. The applications of NLP and deep learning in the field of legal search support have achieved many successes with many outstanding publications. The studies in [6,7,8] use advanced techniques in the field of NLP, such as BERT, GNN and CNN, to understand and process the content of legal documents as well as the content of questions to provide corresponding answers. However, these methods are difficult to implement as they require significant computational resources as well as large amounts of data. Moreover, the ability to handle complex relationships is always difficult with natural language processing techniques.

Currently, prominent LLMs [9,10] likes ChatGPT and Gemini are considered to be a fairly good solution to support legal information search. It is known as a language model that applies deep learning algorithms to perform many NLP tasks [23]. However, a major problem is that LLMs often encounter hallucination and bias. There are currently many unreliable sources of information as well as many anti-state sources that intentionally misrepresent the truth, affecting the quality of answers [25].

And specifically, there are also many research works on question pre-processing to condense the main content of the question as well as remove the system's interference components. The studies in [1,2,18] focused on analyzing Vietnamese Questions to serve the Q&A system in the fields of Education, Law and Social network. Those researches combined techniques in natural language processing (NLP) to create a deep learning model to solve the above problem, the model includes three processing stages: first, take advantage of the PhoBERT model (to represent words with context), next use the bidirectional long-short neural network (BiLSTM) to capture the relationship between words and create a full sentence representation), and finally use the Biaffine classifier to estimate the probability that each pair of start-end words is an important piece of information. The model is trained on a public Vietnamese legal data set, showing that this model achieves an F1-score of up to 94.79%, significantly superior to previous studies.

Ishigaki et al. proposed a method to summarize long questions, to shorten and highlight the main content of the question [3]. It improved the efficiency in optimizing the process of asking and answering questions and searching for information. That method was surveyed and compared with summarization methods. In which, the achievement of the encoder-decoder model combined with the copying mechanism is outstanding results in both ROUGE-2 F-measure and manual evaluation from humans.

However, those studies have not yet analyzed the context of the question in depth and are not suitable for the legal field. This study will present a solution to build a system to support legal information queries with the main goal of the research to solve the cases of misspelling in users' questions as well as combine with the power of LLMs in the process of understanding the context and shortening the question information to improve the system's answering performance.

3. Building a legal query system

3.1. Query system design process

The legal information query system of the research group is built based on two main components, the first component is the knowledge representation platform with the intellectual domain that the system can support searching and the second component is the information query support module based on the previously represented knowledge base simulated in the Figure 1.

With the knowledge representation platform of the system, legal documents in the knowledge domain are organized into knowledge graphs based on the Legal-Onto ontology platform which represents the content of the knowledge domain [17]. The query module component of the system is implemented based on the method of representing questions into knowledge graphs corresponding to the content of the question and finding the matching subgraph in the corresponding knowledge base of the system as the method

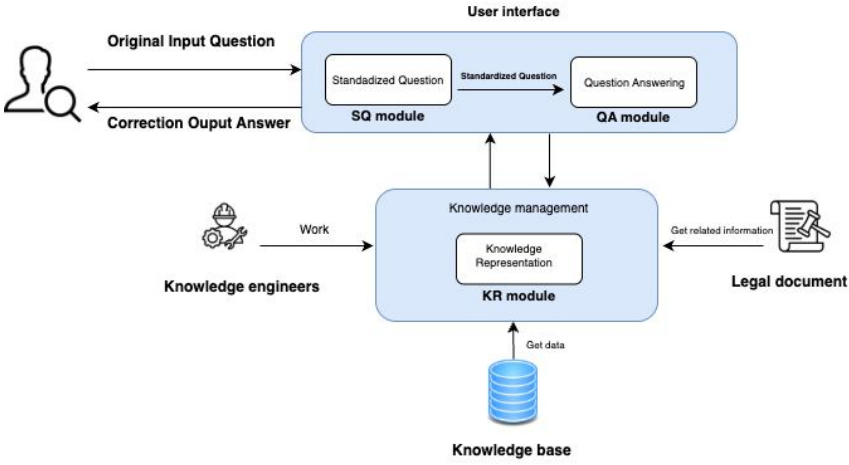


Figure 1. Legal information query system architecture.

presented in the study [24]. And to improve the performance and accuracy in the system's ability to answer questions, the topic standardizes questions with a specific process as in section 3.2.

3.2. Question pre-processing design

In this section, the study presents in detail the techniques for extracting information from questions. First, in section 3.2.1, the study presents the pre-processing method of standardization to help the questions eliminate special characters as well as spelling errors. Next, in section 3.2.2, the process of combining with LLMs to extract information from questions after being standardized is presented.

3.2.1. Question standardization process

The question standardization process ensures that the input questions are converted into a standard form, easy to process and suitable for natural language processing systems (NLP) or AI models. Increase accuracy in classification, information extraction, and question response. In this section, the topic of question pre-processing helps to standardize the questions in turn, which is performed sequentially through the steps of standardizing special characters, then correcting spelling errors and standardizing abbreviations in the question as shown in Figure 2. This process helps the questions become clearer.

Step 1: Standardize special characters and abbreviations

In natural language processing, abbreviations and special characters are often found in question segments on the internet, in casual communication, or in unstructured questions such as user questions. These abbreviations pose many challenges to automatic answering systems when they analyze and extract information from these questions. Standardizing special characters and abbreviations is very important to remove special characters, remove unnecessary elements, and replace them with specific words to make the question semantically clear.

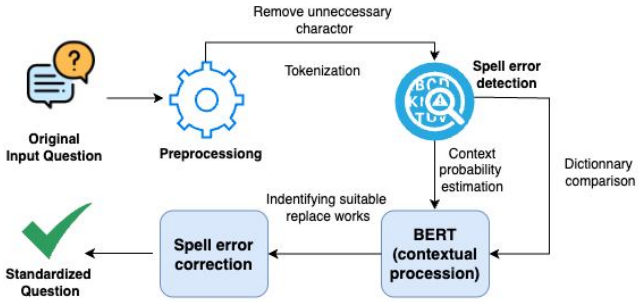


Figure 2. Question standardization pre-processing

To overcome this problem, the research team collected and built a dictionary of abbreviations and special characters including common abbreviations and special characters along with abbreviations and special characters specific to each field as simulated in Figure 3. Along with that, the system will have a self-customization mechanism according to specific fields, in order to improve the accuracy of specialized systems. From there, identify special characters and abbreviations in the question through the built data set and replace them with words that represent the semantics of that symbol.

No.	Type	Ký tự	Normalized Form
1	Special Character	&	và (and)
2	Special Character	/	hoặc (or)
3	Abbreviation	vs	với (with)
4	Abbreviation	BHXH	bảo hiểm xã hội (social insurance)
5	Abbreviation	BHTNDS	bảo hiểm trách nhiệm dân sự (liability insurance)

Figure 3. List of special characters, abbreviations in questions

For example, with the user's question "Điều khiển xe máy nhưng không mang theo BHTNDS & GPLX bị phạt như thế nào?" (What is the penalty for driving a motorbike without carrying a driver's license and insurance?), the system will identify 3 special characters in the sentence:

- BHTNDS: bảo hiểm trách nhiệm dân sự (liability insurance)
- GPLX: giấy phép lái xe (driving license)
- &: và (and)

From there, proceed to standardize and get the complete question result as follows:

- Điều khiển xe máy nhưng không mang theo bảo hiểm trách nhiệm dân sự và giấy phép lái xe sẽ bị phạt như thế nào? (What is the penalty for driving a motorbike without civil liability insurance and a driver's license?)

Step 2: Correcting spelling errors

Detecting and correcting spelling errors is a particularly important step in representing the knowledge of the question. These errors seriously affect the semantic anal-

ysis as well as the extraction of key information in the sentence. To detect and correct spelling errors, this study integrates with a natural language processing tool to identify keyphrases that are not in the dictionary or have a low probability of appearing in the corresponding context. Furthermore, to correct spelling errors accurately, avoid cases of correct corrections becoming incorrect or corrections that change the semantics of the question. The system combines the BERT model to specifically identify contextual information to thereby improve the accuracy in the process of correcting spelling errors, which is simulated as shown in Figure 4.

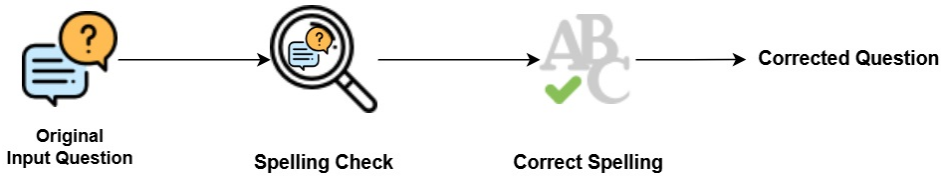


Figure 4. Simulate the process of correcting spelling errors of questions

3.2.2. Extract key information

After the question is pre-processed to remove spelling errors and special characters, the main information is extracted to better understand the context of the question and reduce noise. In this process, the research team takes advantage of the context understanding ability of LLMs to summarize the content of the question, then uses natural language processing tools to extract the main components of the question. This will help the system understand the content of the question better, thereby creating a solid foundation for providing the most accurate answer. The information extraction process is simulated in Figure 5 and specifically in the following steps:

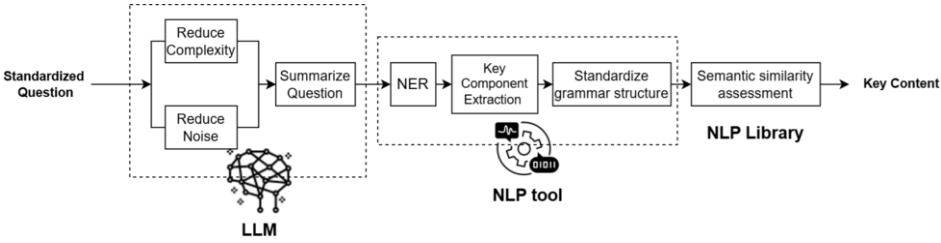


Figure 5. Key information extraction process.

Step 1: Summary of questions

Take advantage of the contextual understanding ability of LLMs. Research using LLMs to summarize the content of the question to eliminate redundant information and retain important information of the question. Helps reduce the complexity of the question as well as reduce the semantics of the main idea of the question. Specifically as follows:
Input: User question after pre-processing for normalization.
Output: The shortened question ensures the following factors:

- Shortened compared to the original question.

- Reduced complexity.
- Reduced semantic noise for the question.

Step 2: Extract entities

The shortened question is extracted to extract important components including entities as well as main verbs in the question to optimize the question analysis process. The process of extracting these important entities is based on the method mentioned in the study [4] and combined with the natural language processing tool PhoNLP to identify the main components in the question as shown in Figure 6. This is to remove secondary elements that do not affect the main meaning of the question.

Input: Question after being shortened.

Output:

- Entities.
- Verbs.
- Additional Information.

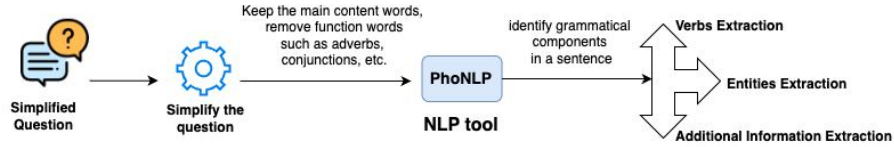


Figure 6. Simulate the process of extracting question entities.

Step 3: Standardize grammar structure

After extracting the entities as well as the main verbs and related information. Restructuring the extracted components is carried out to build a concise question expression, focusing on the core content of the user. The main entities as well as the main verb of the question are brought to the top and the related information is combined and included behind.

Step 4: Semantic similarity assessment

To check the extraction of the above steps, we used the pre-trained BERT model (bert-base-uncased) provided by the Hugging Face Transformers library to encode both the original and the normalized questions into vector representations without additional fine-tuning, We extract the vector representation corresponding to the [CLS] token from each sentence, which is designed to capture the overall meaning of the input. We then compute the cosine similarity between these vectors to assess the semantic similarity between the original and normalized questions.

Specifically, the steps are as follows:

- q_{orig}: Original question.
- q_{ext}: Question after extracting the main content.
- $f : \mathcal{Q} \to \mathbb{R}^d$: Mapping function from question space $\mathcal{Q}$ to feature vector space $\mathbb{R}^d$, implemented by the Sentence Transformer model.

To check the accuracy of extracting the main content of the question, we use Sentence Transformer to map the original question q_{orig} and the extracted question q_{ext} into feature vectors $\mathbf{v}_{\text{orig}}, \mathbf{v}_{\text{ext}} \in \mathbb{R}^d$:

$$\mathbf{v}_{\text{orig}} = f(q_{\text{orig}}), \quad \mathbf{v}_{\text{ext}} = f(q_{\text{ext}}) \tag{1}$$

Next, the cosine similarity between these two vectors is calculated using the formula:

$$\text{Sim}(\mathbf{v}_{\text{orig}}, \mathbf{v}_{\text{ext}}) = \frac{\mathbf{v}_{\text{orig}} \cdot \mathbf{v}_{\text{ext}}}{\|\mathbf{v}_{\text{orig}}\| \|\mathbf{v}_{\text{ext}}\|} \tag{2}$$

This value reflects the degree to which the semantics of the question is preserved after extraction. This process is illustrated in Figure 7.

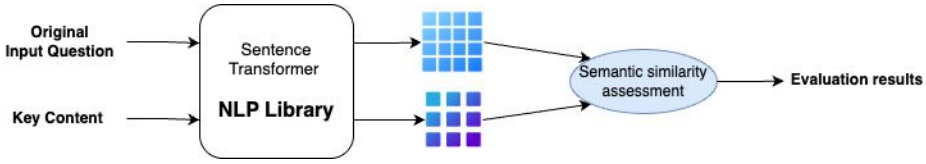


Figure 7. Simulates the process of evaluating the semantic similarity of the reduced question..

3.3. Issues in the question pre-processing process

Question pre-processing plays an essential role in improving the efficiency and increasing the accuracy of QA systems. It is not only the first step of string processing, but pre-processing is also the foundation that helps the system understand the core intention of the question that the user enters. Thereby improving the ability to respond quickly and accurately. A pre-processing process is considered effective when it can eliminate noise factors, minimize unnecessary complexity in each question, and increase the processing speed and accuracy of information retrieval.

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In practice, users often ask questions with natural, flexible language structures that do not follow a fixed pattern. This makes it difficult for the system to understand the user's intention. Therefore, the question-answering system needs to be able to standardize and simplify questions without losing the important semantics of the sentence. Building a question standardization process needs to be designed to ensure accurate extraction of key content, while preserving the original context of the question. This is especially important in situations with ambiguous, unclear factors or when users use reference expressions or linked contexts from previous questions.

In addition, for the pre-processing process to achieve optimal efficiency. It is necessary to build a dataset that supports the normalization of linguistic components such as special characters, abbreviations, common grammatical errors or local slang. This dataset will be combined with large language models (LLMs) and natural language processing (NPL) tools to enhance the ability to analyze and accurately represent the structure of questions.

It is worth noting that although LLMs provide powerful language processing capabilities, they are not completely dependent on them, as LLMs models can generate false or inconsistent information with the context. Therefore, a strong pre-processing layer not only helps normalize the input to match the model's understanding capabilities, but also acts as a semantic filter, helping to stick to the original content of the question and limit the risk of inaccurate responses.

In short, optimizing the pre-process helps improve the performance and reliability of the QA system and better meet the increasing expectations from users. This is an important prerequisite step in building systems that can understand user language more accurately, deeply, and consistently, especially in specialized fields.

4. Experimental results

To verify the effectiveness of this method, the study collected questions that were commented by users on fanpages or legal information pages to test the accuracy of the method. These are questions that are often not standard in terms of structure and spelling in the sentence. From the list of collected questions, the topic extracted information from the questions in turn through the steps of standardizing the questions as presented in section 3.2.1 and the main information extraction section in section 3.2.2. The extracted questions will be entered into the system to clearly see the accuracy of the query improved through the process of extracting the main information of the question.

During the experiment, the research team collected information from reputable online sources about law such as thuvienphapluat, luatvietnam, etc. Specifically, the research team collected and synthesized 70 questions about road traffic law and 71 questions about social insurance. The collected questions were classified into groups. With 70 questions about road traffic law, the research team divided them into 3 main types of questions:

- **Kind 1: Questions about Concepts** - including questions related to definitions and general regulations about road traffic, not related to penalties or specific behaviors of vehicle drivers
- **Kind 2: Questions about Documents, driver's licenses & vehicle regulations** - Including questions related to driver's licenses, vehicle registration, legal documents, and vehicle control rights.
- **Kind 3: Questions on Violations, Penalties** - Includes questions Focusing on fines, violations, deductions and legal handling regulations

As for the 71 questions on the Social Insurance Law, the research team divided them into 4 main types of questions:

- **Kind 1: Questions about concepts** - Includes questions related to the nature, role and scope of application of insurance types (social insurance, health insurance, unemployment insurance, etc.). This is the foundation for people, employees and employers to understand what they are participating in, why they should participate and how to participate.
- **Kind 2: Questions about regimes and benefits** - Includes questions focusing on the specific benefits that insurance participants are entitled to from social insurance, unemployment insurance, and health insurance when eligible. including ma-

ternity, sickness, retirement, death, work accident, occupational disease and unemployment insurance. This is an important group for people to know what they will get when paying social insurance.

- **Kind 3: Questions about contributions, submissions and participation forms** - Includes questions related to how to participate in compulsory and voluntary social insurance, contribution levels and insurance participants, from which to choose the appropriate participation option. This group is very important for freelancers, people working without contracts, or business owners.
- **Kind 4: Questions about documents, procedures and insurance books** - Includes questions related to administrative procedures and insurance records management, issues related to paperwork when leaving work. This group supports people when they need to reissue books, adjust information, look up, complain... to ensure that their rights are not interrupted.

4.1. Evaluate the accuracy of standardized questions

With a total of 141 questions including questions about road traffic laws to protect society. The research team conducted standardization according to the steps mentioned in section 3.1 and performed information extraction as in steps 3.2. To evaluate the degree to which the meaning of the question is preserved after standardization. the research team employed a semantic similarity measurement between the original and the standardized question. Specifically, both questions were encoded using the `BERT model (bert-base-uncased)` provided by the Transformers library from Hugging Face, which maps each sentence into a fixed-size semantic vector. Then, the cosine similarity formula was applied to measure the semantic similarity between the two vectors. This cosine score ranges from 0 (completely dissimilar) to 1 (identical). The average cosine similarity across all pairs of original and The average cosine similarity across all pairs of original and standardized questions was 0.9078, indicating a high level of semantic preservation ($\sim$ 90.78%) and demonstrating the effectiveness of the proposed standardization method. Specifically, BERT model also perform the tokenization of input sentences. This process allows for determining the number of tokens that the BERT model will process. And got positive results with the length of the question measured by the number of tokens obtained after the tokenization process. the original unprocessed question had an average length of 37.11 tokens, and after standardization and reduction, the question had a length of only 19.47 tokens. This reduction of nearly 50% in length—combined with the high semantic similarity.

For example, the initial question entered by the user is ”What is the penalty for driving a two-wheeled motorbike on a road in an urban area, but using a mobile phone?”, then the shortened answer is ”What is the penalty for driving a motorbike while using a mobile phone?”, making the question more concise and clear.

4.2. Evaluate the accuracy of the query

To further evaluate the effectiveness of extracting key information of the question, the topic also compared queries based on 141 collected questions after normalization to perform queries based on the method of representing questions into knowledge graphs and finding the subgraph matching part in the system’s knowledge base as the method pre-

sented in the study [5]. And to evaluate the accuracy of the answer to a question, the research team conducted an assessment based on 3 factors, respectively, the penalty level, additional penalty, and specific basis, with weights showing the importance of the factors presented in the table 1. Because the importance of legal answers is very important, when evaluating the answers, the research team and specifically the experts in the legal field only make right and wrong assessments of the mentioned components.

Table 1. Table of weighting of components in the answer.

Order	Component	Weight
1	Penalty	50%
2	Additional Penalty	40%
3	Basis	10%

In addition, we conducted a comparison between our proposed method and popular large language models (LLMs) such as ChatGPT and Gemini, using the same set of original queries as input. These initial questions were often long, contained spelling errors, and included redundant or ambiguous phrasing—factors that negatively impact the quality of automatic query understanding.

As shown in Tables 2 and 3, our method achieved an average accuracy of 78.55% across all query types, significantly outperforming Gemini under the same input conditions. Notably, for queries related to documents, driver's licenses, and vehicle regulations (Type 2), our approach achieved an accuracy of 82.25%, compared to 39.80% for ChatGPT and 33.12% for Gemini.

It should also be noted that the evaluation results rely in part on manually annotated correctness labels. While our labeling process followed clear criteria—focusing on factual accuracy, completeness of legal references, and alignment with relevant legal articles—it may still carry subjective bias introduced by human annotators. We acknowledge that this labeling approach may affect the objectivity of the evaluation in Tables 2 and 3. To mitigate this limitation, we used consistent annotation guidelines and verified a subset of the labels through discussion among multiple reviewers. In future work, we aim to construct a standardized legal QA dataset with expert-verified gold labels in Vietnamese, to ensure reproducibility and objectivity.

Moreover, our system consistently provides the complete legal basis and associated penalties for each user query, as illustrated in Figure 8.

Table 2. Testing results on Road Traffic Law with the proposed method and LLMs.

Kind	Meaning	Quantity	ChatGPT	Gemini	Proposed method
1	Questions about Concepts	19	74.67%	56.33%	**81.53%**
2	Questions about Documents, driver's licenses & vehicle regulations	22	39.80%	33.12%	**82.25%**
3	Questions on Violations, Penalties	28	64.87%	38.25%	**80.52%**

Table 3. Experimental results of Social Insurance Law using proposed method and LLM.

Kind	Meaning	Quantity	ChatGPT	Gemini	Proposed method
1	Questions about Concepts	6	72.15%	54.60%	**78.30%**
2	Questions about regimes and benefits	27	66.42%	41.75%	**74.88%**
3	Questions about contributions, submissions and participation forms	20	62.73%	35.94%	**76.10%**
4	Questions about documents, procedures and insurance books	18	35.89%	31.20%	**77.80%**

While the comparison results show that the proposed method outperforms ChatGPT and Gemini in noisy query scenarios, it is important to note that the LLMs were evaluated without applying any equivalent pre-processing or fine-tuning on the legal domain. This limitation may affect the fairness of the comparison. Future work will consider integrating the same pre-processing pipeline with LLMs or fine-tuning them on legal-specific datasets to better assess their performance in an optimized setting.

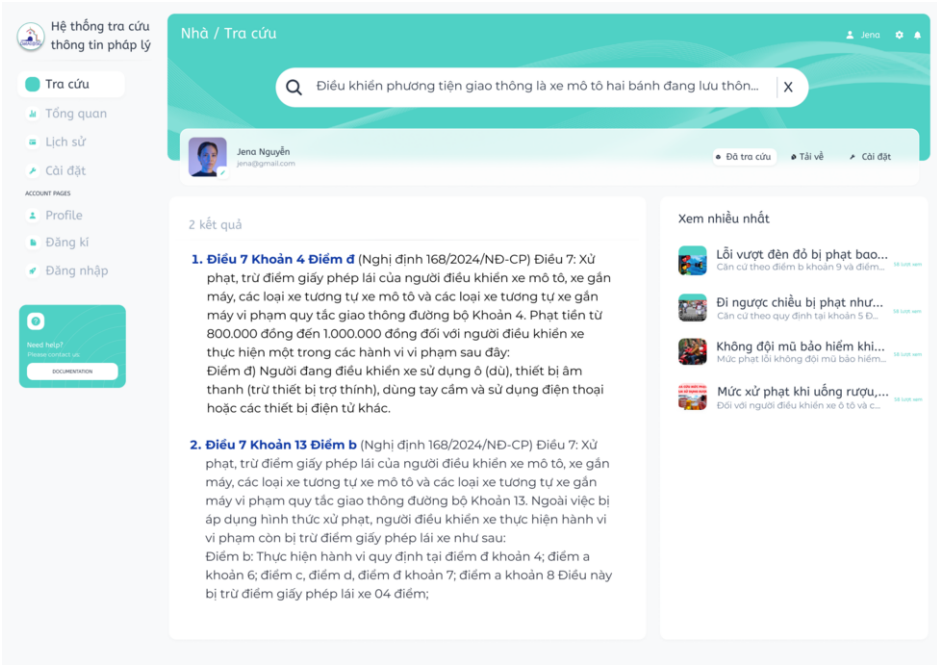


Figure 8. System query interface

5. Conclusions

This method has proven to be effective in answering questions in the legal field. Moreover, the pre-processing method as well as the shortening of the question content ensures that the system can grasp the main meaning of the question posed by the user, avoiding distracting information that affects the query and providing answers. The shortened question based on the context of the original question ensures that the meaning of the original question is preserved. Thereby improving the accuracy rate as well as the performance of the answer. The experimental results demonstrate the feasibility and high accuracy of the method.

In the future, the research team will combine symbolic to perform answer inference as well as identify important logical components of the question that need to be retained to ensure high consistency after shortening. As well as applying more NLP and LLMs tools to reduce human effort as well as improve the accuracy in the process of answering legal questions posed by users.

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